

- gastrula emergent properties, 1066
of hair and keratin, 920
of lens of eye, 1138
levels of biological, 4f–5f
life as emergent property, 125
lignin evolution, 783
in loop of Henle, 998
of neurons, 1084
oxidative phosphorylation and, 186
photosynthesis and shoot architecture, 804
in plant life cycle, 653
sickle-cell disease and, 505
structure and function in, 6
as theme of biology, 2, 3
viral structure and function, 414
of walking and breathing, 756
water as versatile solvent and, 55
- Organizer of Spemann and Mangold, 1062f
- Organ-level herbivore defenses, plant, 868f
- Organ of Corti, **1113f**
- Organogenesis, **1036f**, 1044, **1049**, 1054f–1055f
- Organs, **5f**, **759**, **876t**
digestive. *See* Digestive systems
embryonic germ layers and, 1051f
endocrine system, 1003, 1004f. *See also* Endocrine glands
excretory, 985, 986f–987f
eyes and light-detecting, 1117f–1119f
floral, 823f–824f
human reproductive, 1025f–1027f
immune system rejection of transplanted, 969–970
as level of biological organization, 5f
organogenesis of, 1044, 1049, 1054f–1055f
plant roots, stems, and leaves as, 759f–762f
reverse positioning of human, in situ inversus, 1064f
smooth muscle in vertebrate, 1131–1132
- Organ systems, **876**
internal exchange surfaces of, 875f
mammalian, 876t
- Orgasm, 1034
- Orientation
honeybees, 1142f
leaf, 787
migratory, 1156, 1157f
plant cell expansion, 777f
- Origin-of-life studies, 57f, 58
- Origin of Species, The* (Darwin, C.), 14f, 469, 473, 483–484, 487
- Origins of replication, **242**, 243f, **322**, 323f
- Ornithine, 338f
- Orthologous genes, **565f**, 566
- Orthoptera, 712f
- Oryza sativa*, 438, 509f, 651, 817, 838, 850–851, 1263
- Osculum, **690f**
- Oseltamivir (Tamiflu), 414
- Osmoconformers, **978**
- Osmolarity, **978**, 989, 990f, 991, 998
- Osmoreceptors, 1110
- Osmoregulation, **134**, **978**
challenges and mechanisms of, 978f–980f, 981
energetics of, 980
excretion and. *See* Excretory systems
homeostasis by, 977f, 978
ion movement and gradients and, 993f
osmosis and, 134f–135f
osmosis and osmolarity in, 978
salinity and, 1185f
transport epithelia in, 981, 982f
- Osmoregulators, **978f**
- Osmosis, **134**, **788**, 977
diffusion of water by, across plant plasma membranes, 788, 789f–791f
effects of, on water balance, 133f–135f
osmolarity and, 978
- Osmotic pressure, blood, 933f
- Osteichthyans, **728f**–729f, 730. *See also* Fishes
- Osteoblasts, 878f, 1134
- Osteoclasts, 1134
- Osteons, 878f
- Ostriches, 992f
- Otoliths, 1115f
- Otters, 1189
- Ouabain, 1084
- Outer ear, **1113f**
- Outgroups, **561f**, 572
- Ovalbumin, 76f
- Oval window, **1113f**
- Ovarian cancer, 447
- Ovarian cycle, **1032f**, 1033
- Ovaries, angiosperm, **644f**–646f, 647, **823f**–824f, 830f–831f. *See also* Fruits
- Ovaries, human, 257f, 258, 1004f, 1014, 1015f, **1026**, 1027f
- Overgrazing, 1231
- Overharvesting, 702f, 727–728, 1264, 1265f
- Overnourishment, 917–918
- Overproduction, offspring, 475f, 476
- Oviducts, **1026**, 1027f
- Oviparous organisms, **727**
- Oviraptor* dinosaurs, 564f, 565
- Ovoviviparous organisms, **727**
- Ovulate cones, 640f, 641
- Ovulation, **1021**, 1035f
- Ovules, 636, **638f**, **823f**
- Owl-mouse predation study, 23
- Owls, 920, 1218
- Oxidation, 112, 165, **166f**
- Oxidative fibers, 1130–1131
- Oxidative phosphorylation, 168, **169f**, 174f–177f, 186
- Oxidizing agents, **166f**
- Oxygen
atmospheric, in animal evolution, 677
catabolic pathways and, 165
in circulation and gas exchange, 947f–949f
covalent bonding and, 36, 37f
development of photosynthesis and atmospheric, 532f–534f
diffusion of, across capillary walls, 932
in double circulation, 924f, 925
in ecosystem interaction, 10f, 11
electronegativity of, 37, 45
as essential element, 29f, 64
in gas exchange, 921, 939f, 940
in human breathing, 946
plants production of, 619
in mammalian circulation, 926f
metabolic rate and, 890f
in net ecosystem production, 1242
in organic compounds, 59f
in oxidation, 166f
Permian mass extinction and low levels of, 540
in photosynthesis, 41f
in plant cells, 209f
in plant composition, 809
as product of photosynthesis, 188, 206, 207f
role of, in prokaryotic metabolism, 582
species distributions and availability of, 1185
thyroid hormone and consumption of, 179
transport of, 951
- Oxytocin, 1004f–**1005f**, 1007f, 1037f
- Oystercatcher, 1161
- Oyster drills, 542
- Oysters, 699, 701f, 1020
- Ozone depletion, 1283f, 1284
- P**
- p21* gene, 390
- p53* gene, **389**, 390f–391f
- P680 chlorophyll *a*, 196, 197f
- P700 chlorophyll *a*, 196, 197f, 198
- Pacemaker, heart, 928f
- Pacific Island land snails, 702f
- Pacific salmon (*Oncorhynchus* species), 89
- Pac-man mechanism, 241f
- Paedomorphosis, 544, **545f**, 732
- Paedophryne swiftorum*, 1164f
- Paine, Robert, 1226f
- Pain receptors, **1111**–1112
- Pair bonding, 1144, 1155f
- Pakicetus*, 482f
- Paleoanthropology, **748**
- Paleogene period, 538
- Paleontology, 13f, 88f, **470f**, 471
- Paleozoic era, 530f, 533f, 539f, 677f, 678
- Palisade mesophyll, 771f
- Pallium, avian, 1099f
- Palumbi, S. R., 557f
- Pampas, 1175f
- PAMP-triggered immunity, 866
- Pan-Cancer Atlas, 448
- Pancreas, **909**, 1004f
cellular interactions of, 142
- in digestion, 909
- exocytosis and, 139
- in glucose homeostasis, 916f
- rough ER and, 105
- Pancrustaceans, **708**, 709f–712f, 713
- Pandemics, **410**
- Pandoravirus*, 408
- Pangaea, **482**–483, **539f**, 551
- Pan* genus, 746f. *See also* Chimpanzees
- Panthera pardus*, 554, 555f
- Panthera tigris altaica*, 1288
- Panther chameleons (*Furcifer pardalis*), 735f
- Paper, 651
- Papaya, 757
- Paper wasps, 712f
- Papillae, 1124f
- Papillomaviruses, 394
- Pappochelys*, 737
- Papua New Guinea, 1164f
- Parabasalids, **600f**
- Parabronchi, 944, 945f
- Parachutes, seed and fruit, 832f
- Paracrine signaling, 215f, **1000f**, 1001
- Parahippus*, 548, 549f
- Parakeets, 1136
- Paralogous genes, 565f, **566**
- Paramecium*, 13f, 93f, 134, 135f, 607f, 617, 1199f, 1215
- Paraphyletic groups, **560f**
- Parapodia, 703
- Parasites, 584f, **587**, **1220**
acanthocephalans, 698
animals as, 673f
antigenic variation and, 972
apicomplexans, 605, 606f
arachnids, 708f
cercariae, 609
entamoebas, 613
flatworms, 696f–697f
fungi as, 655, 660, 663, 665, 669f–670f
insects as, 713
lampreys as, 723f, 724
nematodes, 705f, 706
plants as, 818, 819f, 820
protist, 598f, 600f, 614
in zoonotic diseases, 1234
- Parasitism, **587**, **1220**
- Parasitoid wasps, 869f
- Parasympathetic division, peripheral nervous system, 928, 1088f–**1089f**
- Parathion, 158
- Parathyroid glands, 1004f, **1011f**
- Parathyroid hormone (PTH), 1004f, **1011f**
- Parenchyma cells, **764f**, 769f, 770
- Parental alleles, genomic imprinting and, 310f, 311
- Parental care, 1023f, 1150, 1151f, 1200, 1201f–1202f
- Parental types, **302**
- Parietal cells, 907f
- Parietal lobe, 1085, 1097f
- Paris Agreement, 1283
- Paris japonica*, 448t, 449, 843
- Parkinson's disease, 65, 83, 231, 361, 413, 431–432, 671, 1081, **1104**
- Parsimony, 562, 563f, 564
- Parthenogenesis, **697**–698, **1020**, 1021f
- Partial pressure, **939**
- Particulate model of inheritance, 487
- Parus major*, 740f
- Passeriformes, 740f
- Passive immunity, **968**
- Passive transport, 126, **133**
active transport vs., 137f
diffusion as, 132, 133f
facilitated diffusion as, 135f, 136
interpreting scatter plots on glucose uptake as facilitated, 136
in plant cells, 209f
of water across plant plasma membranes, 788, 789f–791f
water balance and osmosis as, 133f–135f
- Patella vulgata*, 547f–548f
- Paternal alleles, 372
- Paternal chromosomes, 265
- Paternity
certainty of, 1150, 1151f
tests for, 437
- Pathogen-associated molecular patterns (PAMPs), **866**